

Table 1. Full HarmBench robust-refusal results for each completed model scale. Values are full-sample means $\pm$ ten-group behavior-clustered jackknife standard errors. Direct requests and the five frozen human-jailbreak templates use matched behavior identities within each model: 240 for Llama-3.2-1B/3B and 120 for the compact Llama-3.1-8B run.

Model	Template	Method	ASR (%) $\downarrow$	Safe-concept relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$	Overall steering (0–2) $\uparrow$
Llama-3.2-1B-Instruct	Direct	Original	10.00 $\pm$ 2.08	1.91 $\pm$ 0.04	1.73 $\pm$ 0.05	1.85 $\pm$ 0.02	1.62 $\pm$ 0.06
		A-LQR	9.58 $\pm$ 1.76	1.95 $\pm$ 0.03	<u>1.51 $\pm$ 0.06</u>	<u>1.89 $\pm$ 0.02</u>	<u>1.46 $\pm$ 0.07</u>
		H_∞ (ours)	2.92 $\pm$ 1.40	1.96 $\pm$ 0.02	1.29 $\pm$ 0.06	1.74 $\pm$ 0.03	1.22 $\pm$ 0.05
	John persona	Original	5.83 $\pm$ 1.67	1.97 $\pm$ 0.02	0.68 $\pm$ 0.09	1.60 $\pm$ 0.06	0.65 $\pm$ 0.08
		A-LQR	6.25 $\pm$ 1.67	1.97 $\pm$ 0.02	<u>0.62 $\pm$ 0.09</u>	<u>1.52 $\pm$ 0.08</u>	<u>0.55 $\pm$ 0.09</u>
		H_∞ (ours)	1.25 $\pm$ 0.89	1.98 $\pm$ 0.01	0.39 $\pm$ 0.07	1.19 $\pm$ 0.07	0.34 $\pm$ 0.06
	DNE nonresponse	Original	6.25 $\pm$ 1.55	1.99 $\pm$ 0.01	0.63 $\pm$ 0.08	1.59 $\pm$ 0.05	0.64 $\pm$ 0.08
		A-LQR	4.17 $\pm$ 1.08	<u>1.99 $\pm$ 0.01</u>	<u>0.57 $\pm$ 0.08</u>	<u>1.53 $\pm$ 0.06</u>	<u>0.56 $\pm$ 0.08</u>
		H_∞ (ours)	1.67 $\pm$ 0.68	1.98 $\pm$ 0.01	0.38 $\pm$ 0.07	1.18 $\pm$ 0.06	0.30 $\pm$ 0.05
	Jailbreak Bot	Original	4.58 $\pm$ 1.45	1.96 $\pm$ 0.03	0.64 $\pm$ 0.09	1.62 $\pm$ 0.06	0.63 $\pm$ 0.09
		A-LQR	3.33 $\pm$ 1.50	1.96 $\pm$ 0.02	<u>0.56 $\pm$ 0.09</u>	<u>1.53 $\pm$ 0.07</u>	<u>0.53 $\pm$ 0.10</u>
		H_∞ (ours)	0.42 $\pm$ 0.42	1.98 $\pm$ 0.01	0.38 $\pm$ 0.07	1.17 $\pm$ 0.05	0.30 $\pm$ 0.06
	YOJA/Nona roleplay	Original	8.33 $\pm$ 1.08	1.97 $\pm$ 0.02	0.63 $\pm$ 0.08	1.56 $\pm$ 0.05	0.62 $\pm$ 0.09
		A-LQR	8.75 $\pm$ 1.45	1.97 $\pm$ 0.02	<u>0.55 $\pm$ 0.08</u>	<u>1.50 $\pm$ 0.08</u>	<u>0.51 $\pm$ 0.09</u>
		H_∞ (ours)	0.83 $\pm$ 0.56	1.98 $\pm$ 0.01	0.37 $\pm$ 0.07	1.15 $\pm$ 0.07	0.29 $\pm$ 0.05
	APM programmer	Original	2.92 $\pm$ 0.89	1.94 $\pm$ 0.03	0.64 $\pm$ 0.08	1.56 $\pm$ 0.06	0.64 $\pm$ 0.08
		A-LQR	5.00 $\pm$ 1.50	1.97 $\pm$ 0.03	<u>0.57 $\pm$ 0.08</u>	<u>1.51 $\pm$ 0.07</u>	<u>0.53 $\pm$ 0.08</u>
		H_∞ (ours)	0.42 $\pm$ 0.42	1.98 $\pm$ 0.01	0.37 $\pm$ 0.07	1.12 $\pm$ 0.06	0.30 $\pm$ 0.06
Llama-3.2-3B-Instruct	Direct	Original	16.67 $\pm$ 2.48	1.88 $\pm$ 0.05	1.21 $\pm$ 0.06	1.88 $\pm$ 0.03	1.19 $\pm$ 0.06
		A-LQR	20.00 $\pm$ 2.90	1.91 $\pm$ 0.02	1.18 $\pm$ 0.07	1.86 $\pm$ 0.02	1.15 $\pm$ 0.07
		H_∞ (ours)	8.33 $\pm$ 1.86	1.99 $\pm$ 0.01	1.19 $\pm$ 0.06	1.90 $\pm$ 0.02	1.18 $\pm$ 0.06
	John persona	Original	11.67 $\pm$ 2.31	1.90 $\pm$ 0.03	0.61 $\pm$ 0.07	1.54 $\pm$ 0.08	0.58 $\pm$ 0.08
		A-LQR	11.67 $\pm$ 2.69	<u>1.96 $\pm$ 0.02</u>	<u>0.69 $\pm$ 0.08</u>	1.52 $\pm$ 0.06	<u>0.63 $\pm$ 0.07</u>
		H_∞ (ours)	4.17 $\pm$ 1.08	1.94 $\pm$ 0.02	0.60 $\pm$ 0.08	1.65 $\pm$ 0.05	0.59 $\pm$ 0.08
	DNE nonresponse	Original	10.83 $\pm$ 2.08	1.93 $\pm$ 0.02	0.57 $\pm$ 0.07	1.52 $\pm$ 0.07	0.55 $\pm$ 0.07
		A-LQR	10.42 $\pm$ 2.17	<u>1.95 $\pm$ 0.03</u>	<u>0.63 $\pm$ 0.06</u>	1.52 $\pm$ 0.06	<u>0.59 $\pm$ 0.06</u>
		H_∞ (ours)	3.75 $\pm$ 0.97	1.95 $\pm$ 0.02	0.59 $\pm$ 0.08	1.66 $\pm$ 0.05	0.58 $\pm$ 0.08
	Jailbreak Bot	Original	14.17 $\pm$ 2.17	1.90 $\pm$ 0.02	0.57 $\pm$ 0.08	1.54 $\pm$ 0.08	0.56 $\pm$ 0.08
		A-LQR	12.92 $\pm$ 1.70	<u>1.97 $\pm$ 0.01</u>	<u>0.64 $\pm$ 0.09</u>	1.51 $\pm$ 0.06	<u>0.60 $\pm$ 0.09</u>
		H_∞ (ours)	3.75 $\pm$ 1.15	1.93 $\pm$ 0.02	0.59 $\pm$ 0.08	1.67 $\pm$ 0.05	0.58 $\pm$ 0.08
	YOJA/Nona roleplay	Original	11.25 $\pm$ 1.53	1.91 $\pm$ 0.03	0.59 $\pm$ 0.08	1.52 $\pm$ 0.08	0.55 $\pm$ 0.07
		A-LQR	11.67 $\pm$ 1.04	<u>1.95 $\pm$ 0.02</u>	<u>0.65 $\pm$ 0.08</u>	1.51 $\pm$ 0.07	<u>0.59 $\pm$ 0.08</u>
		H_∞ (ours)	4.17 $\pm$ 0.88	1.94 $\pm$ 0.02	0.56 $\pm$ 0.08	1.64 $\pm$ 0.04	0.56 $\pm$ 0.08
	APM programmer	Original	10.83 $\pm$ 2.26	1.91 $\pm$ 0.03	0.56 $\pm$ 0.08	1.48 $\pm$ 0.07	0.53 $\pm$ 0.07
		A-LQR	11.25 $\pm$ 2.49	<u>1.97 $\pm$ 0.02</u>	<u>0.65 $\pm$ 0.07</u>	1.47 $\pm$ 0.07	<u>0.60 $\pm$ 0.07</u>
		H_∞ (ours)	2.50 $\pm$ 1.11	1.96 $\pm$ 0.02	0.58 $\pm$ 0.08	1.60 $\pm$ 0.06	0.58 $\pm$ 0.08

Table 1. Full HarmBench robust-refusal results (continued).

Model	Template	Method	ASR (%) ↓	Safe-concept relevance (0–2) ↑	Instruction relevance (0–2) ↑	Fluency (0–2) ↑	Overall steering (0–2) ↑
Llama-3.1-8B-Instruct	Direct	Original	15.83 ± 4.02	1.83 ± 0.06	1.06 ± 0.07	1.73 ± 0.05	0.98 ± 0.07
		A-LQR	15.83 ± 3.82	1.76 ± 0.05	<u>1.32 ± 0.08</u>	<u>1.77 ± 0.04</u>	<u>1.24 ± 0.09</u>
		H_∞ (ours)	3.33 ± 1.84	1.94 ± 0.03	0.99 ± 0.08	1.76 ± 0.04	1.00 ± 0.08
	John persona	Original	23.33 ± 2.08	1.80 ± 0.05	0.52 ± 0.07	1.39 ± 0.09	0.37 ± 0.08
		A-LQR	15.00 ± 3.24	1.87 ± 0.03	<u>0.68 ± 0.10</u>	<u>1.44 ± 0.07</u>	<u>0.60 ± 0.09</u>
		H_∞ (ours)	1.67 ± 1.11	1.98 ± 0.02	0.31 ± 0.07	0.87 ± 0.06	0.31 ± 0.07
	DNE nonresponse	Original	14.17 ± 4.31	1.83 ± 0.07	0.42 ± 0.07	1.43 ± 0.08	0.34 ± 0.08
		A-LQR	11.67 ± 4.68	1.88 ± 0.04	<u>0.58 ± 0.09</u>	<u>1.46 ± 0.07</u>	<u>0.54 ± 0.10</u>
		H_∞ (ours)	0.00 ± 0.00	1.98 ± 0.02	0.33 ± 0.07	0.90 ± 0.08	0.32 ± 0.07
	Jailbreak Bot	Original	9.17 ± 2.62	1.78 ± 0.07	0.44 ± 0.08	1.39 ± 0.11	0.32 ± 0.08
		A-LQR	7.50 ± 2.62	1.87 ± 0.04	<u>0.58 ± 0.11</u>	<u>1.48 ± 0.06</u>	<u>0.55 ± 0.10</u>
		H_∞ (ours)	0.00 ± 0.00	1.95 ± 0.04	0.31 ± 0.07	0.88 ± 0.10	0.28 ± 0.08
	YOJA/Nona roleplay	Original	15.00 ± 4.78	1.77 ± 0.07	0.45 ± 0.06	1.49 ± 0.09	0.33 ± 0.07
		A-LQR	9.17 ± 3.15	1.86 ± 0.06	<u>0.67 ± 0.08</u>	<u>1.46 ± 0.09</u>	<u>0.63 ± 0.08</u>
		H_∞ (ours)	0.00 ± 0.00	1.95 ± 0.04	0.29 ± 0.07	0.88 ± 0.09	0.28 ± 0.07
	APM programmer	Original	11.67 ± 3.56	1.77 ± 0.06	0.42 ± 0.10	1.38 ± 0.11	0.34 ± 0.08
		A-LQR	10.83 ± 3.06	1.85 ± 0.04	<u>0.62 ± 0.09</u>	<u>1.41 ± 0.08</u>	<u>0.55 ± 0.10</u>
		H_∞ (ours)	0.83 ± 0.83	1.98 ± 0.02	0.30 ± 0.07	0.88 ± 0.09	0.28 ± 0.07